



NE235 – Nuclear Reactor Operations

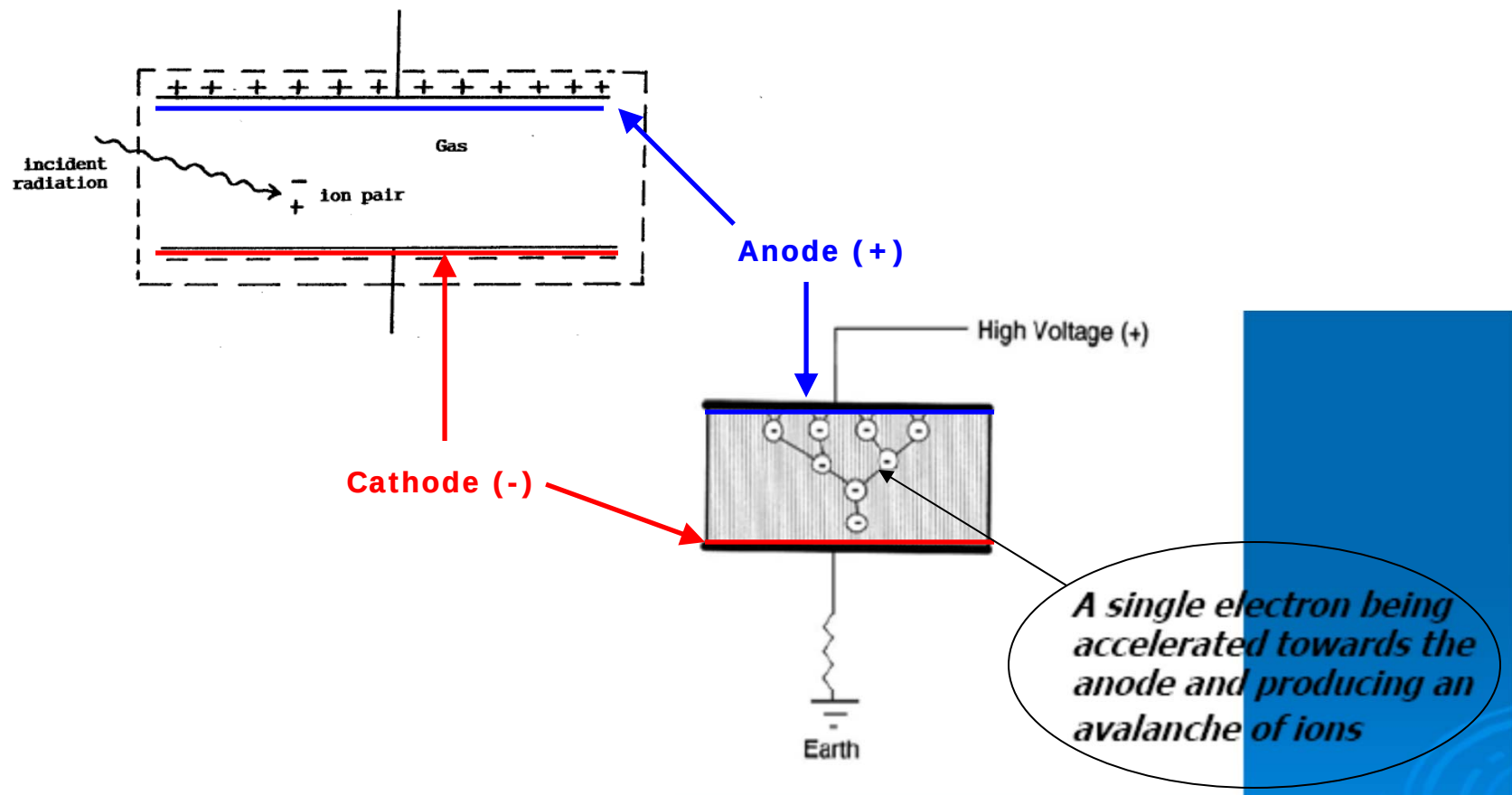
Lab Module #2 Health Physics (HP) Instruments & Dosimetry

Objectives:

- Review Radiation Detection and Measurement – how α^{2+} , β^{-} , γ , and neutrons interact in HP survey instruments.
- Learn to correctly operate and take readings from:
 - Geiger Muller Detectors
 - Ion Chambers
 - Neutron “REM Ball” Monitor
- Review types of Radiation Dosimetry issued to personnel
- Perform an Area Radiation Survey of the PULSTAR Reactor

Radiation Detection and Measurement

Basic configuration of gas or air filled radiation detectors:

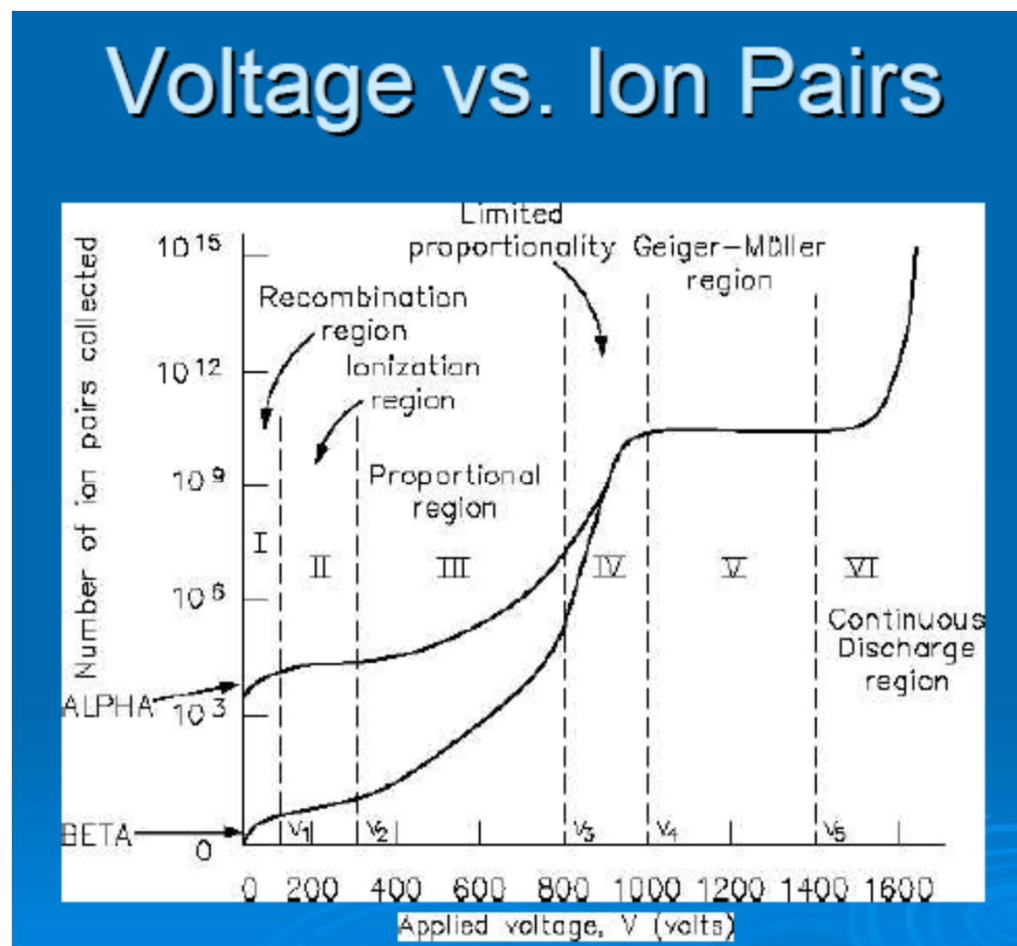


Radiation Detection and Measurement

Amount of positive voltage bias applied to Anode determines number of ion pairs created per ionization event in gas filled detector.

+100-800V = 'Ionization / Proportional Region' where number of ion-pairs produced is proportional to amount of energy deposited.

+1000-1400V = "Geiger-Muller Region" where any ionization event causes avalanche cascade yielding a high amplitude pulse. This pulse is not proportional to energy deposited.



GM Detectors

- ❑ A fill gas (typically a mixture of Argon and Methane) is contained in a sealed tube of various shapes, sizes
- ❑ Ionization occurs in the fill gas and in the detector wall material (wall is at ground potential and serves as the cathode)
- ❑ GM detectors produce a full amplitude pulse (count) for any ion that enters the detector (no energy distinction).
- ❑ Used to detect alpha, beta, gamma and x-rays
- ❑ Used primarily for contamination surveys, reading out as a count rate (CPM).
- ❑ May be calibrated to give **approximate exposure rate** (mR/hr) information for specific type of radiation field.



Ion Chambers

- ❑ Fill gas is air that may or may not be contained in a sealed chamber
- ❑ Ionization occurs in the fill gas and in the detector wall material (wall is at ground potential and serves as the cathode)
- ❑ Ions in the fill gas are collected at the detector anode and produce a pulse height proportional to the number of interactions, or ionization, occurring in the detection volume
- ❑ Ion chambers are very accurate and allow distinction between radiation types based on pulse height.
- ❑ Used to detect beta, gamma and x-rays
- ❑ Used to **accurately measure exposure rate** (mR/hr) for external radiation fields.



Scintillation Detectors

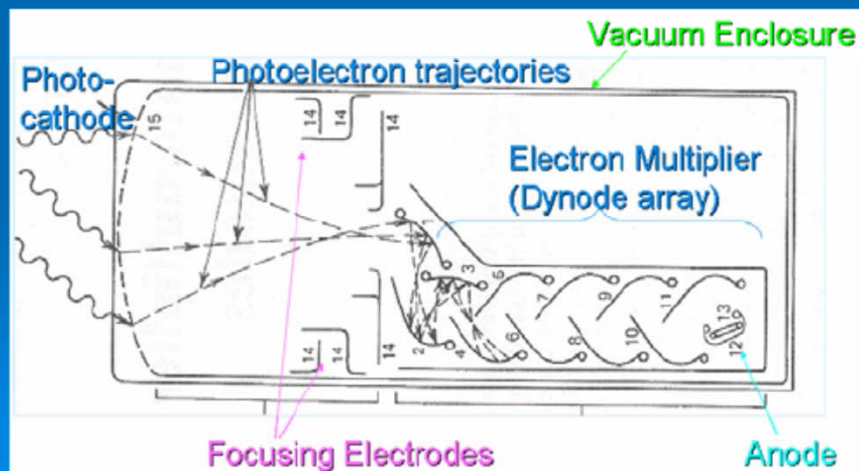


Scintillation crystals respond to radiation interactions by producing light. The light is then detected (e.g. via a Photo Multiplier Tube - PMT) and converted to an electrical signal with a pulse height proportional to the radiation energy deposition rate.

Scintillators are extremely sensitive to radiation and are often used for obtaining energy spectra.

Photomultiplier tubes (PMTs)

- Converts light photons into electrical pluses and amplifies them.
- The amplitude of the pulse is proportional to the amount of light the PMT received.



Neutron Detectors

- ❑ Neutrons don't interact with electrons, so Boron (^{10}B) or Lithium (^6Li) are utilized due to their large thermal neutron absorption cross sections.
- ❑ Neutron is captured by a Lithium or Boron nucleus, ejecting an alpha particle which causes ionization in the detector.
- ❑ Our Neutron 'REM Ball' utilizes a BF_3 (Boron Tri-Fluoride) gas filled tube detector.
- ❑ The polyethylene ball moderates high energy neutrons through scattering, causing them to lose energy and increase their probability of being detected in the BF_3 tube.
- ❑ High energy neutrons have a higher 'Quality factor' (or RBE factor) than thermal neutrons, so result in a higher dose relative to exposure.
- ❑ Reads out in units of mREM/hr.
- ❑ Utilized for performing surveys of external neutron fields around neutron sources and beamport experiments.



Radiation Dosimetry

LANDAUER®

Luxel®+ Dosimeter for X, Gamma, Beta, and Neutron Radiation



Radiations Measured	Photon (X and Gamma Ray)	Beta Particle	Neutron
Detector	Al ₂ O ₃ (Aluminum Oxide)	Al ₂ O ₃ (Aluminum Oxide)	Optional Neutrak® 144 detector inside dosimeter (CR-39)
Analysis Method	Optically Stimulated Luminescence (OSL)	Optically Stimulated Luminescence (OSL)	Chemical etching followed by track counting (Track-Etch®)
Energies Detected	5 keV to in excess of 40 MeV	150 keV to in excess of 10 MeV (Expressed as Average Energy)	Fast: 40 keV to 40 MeV Thermal/ Intermediate: 0.25 eV to 40 keV
Dose Measurement Range	1 mrem to 1000 rem (10 µSv to 10 Sv)	10 mrem to 1000 rem (100 µSv to 10 Sv)	Fast: 20 mrem to 25 rem (200 µSv to 250 mSv) Thermal/ Intermediate: 10 mrem to 5 rem (100 µSv to 50 mSv)
Accuracy	Deep Dose (Hp10) ±15% at the 95% confidence interval for photons above 20 keV Shallow Dose (Hp 0.07) ±15% at the 95% confidence interval for photons above 20 keV and beta particles above 200 keV		
Accreditations, Approvals, Licenses	NVLAP (NVLAP Lab Code 100518-0) for Whole Body (ANSI HPS N13.11-2001) in the comprehensive subcategory "General" in all categories including VI when neutron component is added; and for extremity (ANSI HPS N13.32-1995). HSE (Health and Safety Executive) United Kingdom approved for Whole Body (OSL) and Whole Body Neutrons. DOELAP (Department of Energy Laboratory Accreditation Program).		

Landauer 'Luxel+' Dosimetry

Luxel+ and OSL Technology

Landauer grows the specially formulated aluminum oxide ($\text{Al}_2\text{O}_3\text{:C}$) crystalline detector material. The Al_2O_3 detector is then configured into a thin strip sandwiched within a multi-element filter pack. The filter pack is heat sealed within a laminated, light-tight paper wrapper creating an integrated, self-contained packet that is RF (radio-frequency) sealed inside a tamper-proof plastic blister pack to eliminate possible mishandling, light leakage, or lost detection elements.

Luxel+ may be used for up to one year. It is unaffected by heat, moisture, and pressure when the clear blister pack is uncompromised.

Radiation exposure is measured in Landauer's laboratory by stimulating the Al_2O_3 material with selected frequencies of laser light causing it to luminesce in proportion to the amount of radiation exposure. The luminescence measured is applied to a dose algorithm that relies on the response ratios between different filter positions within the dosimeter to discriminate between beta and photon (x and gamma) radiation fields to determine exposure results.

Dose equivalents arising from exposures to photons (x or gamma rays) will have a deep, lens of eye and shallow value reported. Depending on the energy of the x or gamma rays, these values may or may not be equal. Beta exposures are reported only as a shallow dose equivalent.

Luxel+ and Track Etch Technology

The optional Neutrak detector available for an additional charge is a CR-39 (allyl diglycol carbonate) based, solid-state nuclear track detector that measures exposure due to neutrons. It is not sensitive to x, beta or gamma radiation, and is sealed inside the Luxel+ plastic blister pack to eliminate possible mishandling or lost detection elements. The CR-39 is laser engraved for permanent identification to assure chain-of-custody.

The fast neutron option uses a polyethylene radiator for fast neutrons that records recoil protons resulting from neutron interactions in the dosimeter. The thermal/ intermediate neutron option has a dosimeter design intended for fast, intermediate, and thermal neutrons. The left area of the chip uses a polyethylene radiator for fast neutrons while the right area uses a boron loaded Teflon® radiator fast, intermediate, and thermal neutrons that records alpha particles resulting from neutron interactions in the dosimeter.

During analysis in our laboratory, the CR-39 is etched for 15 hours in a chemical bath to enlarge exposure tracks. The fast neutron dose is measured by counting the tracks generated as a result of the proton recoil with the polyethylene radiator, while the thermal/ intermediate dose is measured by counting the alpha tracks generated with the boron radiator.

Exercise: Operation of Instruments

Pre-Operational Instrument Checks:

- ❑ **Perform a battery check prior to using the detector.**
- ❑ **Check the calibration sticker to be sure the detector is within specification.**

NOTE: Be careful of thin end windows on ion chambers and GM tubes – they are easily damaged.

PULSTAR Lab HP Instruments:

- ❑ Ludlum Model 3 GM w/ Pancake Probe; readout in CPM. Sensitive to $\alpha/\beta/\gamma$
- ❑ Ludlum Model 3 GM w/ Tube; readout in mR/hr. Sensitive to $\alpha/\beta/\gamma$
- ❑ Ludlum 19 'Micro-R' (NaI Scintillator); readout in μ R/hr. Sensitive to γ only
- ❑ Ludlum 12-4 Count Ratemeter Neutron REM Ball; readout in mREM/hr. Sensitive to neutrons only.

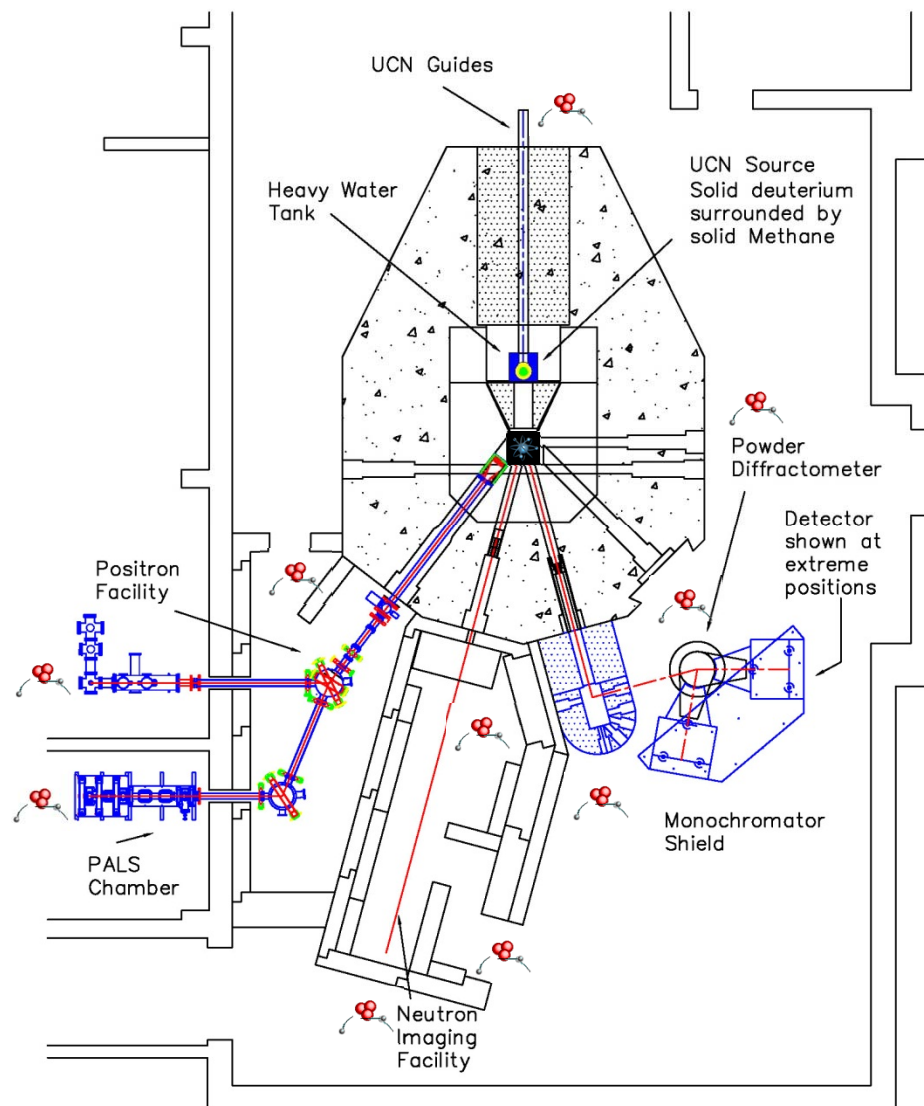
Radioactive Check Sources:

- ❑ Cesium-137: β 's + 661 keV γ 's
- ❑ Cobalt-60: β 's + 1173 & 1332 keV γ 's
- ❑ Beta emitters: Tc-99, Cl-36, Bi/Pb-210, Sr/Y-90
- ❑ Alpha emitter: Pu-239!
- ❑ No neutron source – sorry!

Shielding Materials:

- ❑ Aluminum beta absorbers
- ❑ Lead sheets
- ❑ Paper
- ❑ Plastic

PULSTAR Area Radiation Survey



USE ION CHAMBERS AND NEUTRON REM BALL TO MEASURE RADIATION FIELDS AROUND THE PULSTAR BIOSHIELD AND BEAMPORT EXPERIMENTS.

SURVEY:

- ☐ POOLTOP
- ☐ REACTOR BIOSHIELD
- ☐ BEAMPORTS
- ☐ EXPERIMENTAL FACILITIES
- ☐ BEAMSTOPS
- ☐ AROUND PENETRATIONS THROUGH SHIELDS



RADIATION AREA:

≥ 5 MREM/HR AT 30 CM FROM SOURCE OR SURFACE OF SHIELD.



HIGH RADIATION AREA:

≥ 100 MREM/HR AT 30 CM FROM SOURCE OR SURFACE OF SHIELD.
RWP REQUIRED FOR ENTRY!!